Nicolas Zilberstein

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https://sites.google.com/view/nzilberstein — https://github.com/nzilberstein

EDUCATION

Rice University — Houston, TX
Ph.D. in Electrical and Computer Engineering — 2021 - 2026
Advisor: Prof. Santiago Segarra
Keywords: Inverse problems, Diffusion models, Sampling, Computational imaging

Rice University — Houston, TX
Master of Science in Electrical and Computer Engineering — 2021 - 2023
Thesis: Annealed Langevin dynamics for massive MIMO communications

University of Buenos Aires — Buenos Aires, Argentina
B.S. in Electrical Engineering (with honors) — 2013 - 2020
Thesis: Development of inference algorithms for UWB-GPR signals

WORK EXPERIENCE

Google — Mountain View, CA
Student Researcher at Computational Imaging Team (LUMA) — May 2026 - August 2026
PIs: Kuldeep Purohit and Mojtaba Sahraee-Ardakan
Keywords: Diffusion models, Inference-time scaling, Inverse problems.

Center for Computational Neuroscience, Flatiron Institute — New York City, NY
Summer research intern at the Computational Vision group — May 2025 - August 2025
PIs: Eero Simoncelli and Florentin Guth
Keywords: Energy-based model, inverse problems, anisotropic diffusion models.

InterDigital — Los Altos, CA
Research intern - AI for wireless — May 2024 - August 2024
PIs: Akshay Malhotra and Shahab Hamidi-Rad
Keywords: Topological autoencoder, zero-shot stitching, latent space communication.

ExxonMobil — Buenos Aires, Argentina
Data Engineer — 2019 - 2021
Keywords: Front and Back-end, Cloud computing, Agile methodologies.

School of Engineering, University of Buenos Aires — Buenos Aires, Argentina
Research Assistant — 2018 - 2019
PIs: Prof. Bruno Cernuschi-Frias, Prof. Juan Augusto Maya and Prof. Andres Altieri
Keywords: Particle filter, ground penetrating radar imaging, Green's function, sparse Bayesian learning.

AWARDS

Grants

NVIDIA Academic Grant — 2026

AI generated visuals: a solution for image demand in social science research — Rice University
Center for Computational Insights on Inequality and Society (5000 USD) — 2025

Fellowships

Ken Kennedy – HPE Cray Graduate Fellowship — Rice University
One-year graduate fellowships for outstanding graduate students at Rice University — 2024

UBACyT Scholarship (declined) — University of Buenos Aires
Scholarship for undergraduate students to develop a research project — 2018

Honors

Top reviewer NeurIPS
Selected as a top 8% reviewer based on the area chairs rating 2026

Top reviewer (Silver) ICML
Selected as a top reviewer based on the area chairs rating 2026

Google PhD fellowship nominee Rice University
One of two nominees selected by the ECE department 2025

Excellence in Research Rice University
Award from the ECE section of the Gulf Coast Undergraduate Research Symposium (GCURS) 2020

Travel grants

ICML Travel Grant 2026
ICLR, AISTATS Travel Grant 2025
REA (Rice Engineering Alumni) Graduate Student Travel Grant 2025

PUBLICATIONS

Journals

[J1]
[J2]
[J3]
[J4]

Conference Papers

[C1]
[C2]
[C3]
[C4]
[C5]
[C6]
[C7]
[C8]
[C9]
[C10]
[C11]
[C12]
[C13]

Workshop Papers

[W1]
[W2]
[W3]
[W4]

Patents

1. "Systems and methods for statistics based and topology based interoperable AI/ML model monitoring" 2024
   Akshay Malhotra, Yugeswar Deenoo, Nicolas Zilberstein, Shahab Hamidi-Rad, Mohamed Salah Ibrahim
   InterDigital, US Patent Ref. 18889096

Preprints

[P1]

*Equal contribution

TEACHING EXPERIENCE

Department of Electrical and Computer Engineering, Rice University Houston, TX
Teaching Assistant

- ELEC 573 Network Science and Analytics, *Graduate course* Fall 2022, 2024
- ELEC 301 Signals and Systems, *Undergraduate course* Spring 2022, 2023, 2024, 2025

Graduate school of Computer Engineering, University of Buenos Aires Buenos Aires, Argentina
Teaching Assistant

- Deep Learning Fall 2021

Department of Physics, University of Buenos Aires Buenos Aires, Argentina
Teaching Assistant

- 82.02: Physics II (Electromagnetism) Spring 2020, Fall 2020

Department of Electronic Engineering, University of Buenos Aires Buenos Aires, Argentina
Teaching Assistant

- 86.55: Detection and Estimation theory Spring 2018, 2019

PROFESSIONAL ACTIVITIES

Invited talks

From Guided Sampling to Learned Solvers: Diffusion Models for Inverse Problems Dec 2025
Universidad de la Republica Uruguay

From Guided Sampling to Learned Solvers: Diffusion Models for Inverse Problems Nov 2025
Yusu Wang's group at *UCSD*

Towards fast samplers for solving inverse problems using diffusion models June 2024
Pendulum Systems

Organization

Special Session: Generative AI for Graphs Asilomar 2026
Organizer and Technical Chair

Workshops

Advances of Approximate Bayesian Inference (AABI) 2025 Singapore
Repulsive Latent Score Distillation for Solving Inverse Problems **?** presented in the fast track

Khipu AI 2025 Santiago, Chile
The largest meet-up gathering students, researchers, and innovators in the field of Artificial Intelligence and Machine Learning across Latin America

LOGML Summer School 2024 London, UK
Workshop that aims to bring together mathematicians and computer scientists to work together on a variety of problems at the intersection of geometry and machine learning

Graph Signal Processing Workshop 2023 Oxford, UK
Workshop on graph signal processing. Poster presentation.

Reviewer

Journals: IEEE Comm. Letter; IEEE Trans. Cogn. Commun.; IEEE Trans. Signal Process.; IEEE Trans. Commun.
Conferences: ICCASP, EUSIPCO, ICC, SAM, CAMSAP, NeurIPS (2024, 2025), ICLR (2026), ICML (2026)

Memberships

IEEE Member 2021 - Present
IEEE Signal Processing Society Member 2021 - Present

STUDENT ADVISEMENT

Graduate

Ali Azizpour

Undergraduate

Haoming Chen	Summer 2025
Uros Males, Mila Bokan	Summer 2023

OTHER SKILLS

Languages: Spanish (native), English (fluent)

COMMUNITY SERVICES

Evaluator at Gulf Coast Undergraduate Research Symposium (GCURS)	2024, 2025
Mentor of incoming ECE PhD students - Data science track	2024
Organizer, LatAm RICE Networks, Robots, and ML Minisymposium, Rice Univ.	2021
Member of the Council of the Faculty of Engineering, Univ. of Buenos Aires	2019 - 2021
Member, Advisory Board, Dept. of Electronics, Univ. of Buenos Aires	2018 - 2021
Member, Curricular Committee, Electronic Engineering Degree, Univ. of Buenos Aires	2017 - 2021